but this is nothing compared to the glorious genetic possibilities that open up before anyone specializing in sons. Therefore genes for producing sons will tend to become more numerous, and the pendulum will swing back.

For simplicity I have talked in terms of a pendulum swing. In practice the pendulum would never have been allowed to swing that far in the direction of female domination, because the pressure to have sons would have started to push it back as soon as the sex ratio became unequal. The strategy of producing equal numbers of sons and daughters is an evolutionarily stable strategy, in the sense that any gene for departing from it makes a net loss.

I have told the story in terms of numbers of sons versus numbers of daughters. This is to make it simple, but strictly it should be worked out in terms of parental investment, meaning all the food and other resources that a parent has to offer, measured in the way discussed in the previous chapter. Parents should *invest* equally in sons and daughters. This usually means they should have numerically as many sons as they have daughters. But there could be unequal sex ratios that were evolutionarily stable, provided correspondingly unequal amounts of resources were invested in sons and daughters. In the case of the elephant seals, a policy of having three times as many daughters as sons, but of making each son a supermale by investing three times as much food and other resources in him, could be stable. By investing more food in a son and making him big and strong, a parent might increase his chances of winning the supreme prize of a harem. But this is a special case. Normally the amount invested in each son will roughly equal the amount invested in each daughter, and the sex ratio, in terms of numbers, is usually one to one.

In its long journey down the generations therefore, an average gene will spend approximately half its time sitting in male bodies, and the other half sitting in female bodies. Some gene effects show themselves only in bodies of one sex. These are called sex-limited gene effects. A gene controlling penis-length expresses this effect only in male bodies, but it is carried about in female bodies too and may have some quite different effect on female bodies. There is no reason why a man should not inherit a tendency to develop a long penis from his mother.

In whichever of the two sorts of body it finds itself, we can expect a gene to make the best use of the opportunities offered by that sort of